1. Algebra & Graphs

1.1. Algebraic Manipulation

Algebraic Simplification

- To simplify, combine terms that have the same variable and exponent.

Q1. $2a + 3b + 5a - 9b$

Step 1: Group like terms as they have the same variable raised to the same power

- $(2a + 5a) + (3b - 9b)$

Step 2: Simplify each group and write the simplified expression

- $\boxed{7a - 6b}$

Algebraic Expansion

- When expanding products, multiply the values outside the brackets by the values inside the brackets.

Q1. $3x(2x - 4y)$

Step 1: Apply Distributive Law

- $a(b + c) = ab + ac$
- $3x(2x) + 3x(-4y)$

Step 2: Multiply each term inside by 3x

- $(3x)(2x) = 6x^2$
- $(3x)(-4y) = -12xy$

Step 3: Write the result

- $\boxed{6x^2 - 12xy}$

Q2. $(3x - 5)(4x + 8)$

Step 1: Distribute $3x$ across the second bracket

- $3x \times 4x = 12x^2$
- $3x \times 8 = 24x$

Step 2: Distribute -5 across the second bracket

- $-5 \times 4x = -20x$
- $-5 \times 8 = -40$

Step 3: Combine all terms

- $12x^2 + 24x - 20x - 40$

Step 4: Simplify like terms

- $24x - 20x = 4x$

Step 5: Write the results

- $\boxed{12x^2 + 4x - 40}$

Algebraic Factorization

Q1. Factorize $12x^2 + 18x$

Step 1: Find the HCF of the terms

- HCF of coefficients 12 and 18 is 6.
- HCF of variables x^2 and x is x.
- Overall HCF is $6x$

Step 2: Factor out the HCF

- Write the HCF outside a bracket and divide each term by $6x$.
- $12x^2 \div 6x = 2x$, $18x \div 6x = 3$

Step 3: Write the results

- $\boxed{6x(2x + 3)}$

Completing the Square

$$x^2 + bx + c$$
$$=$$
$$\left(x + \frac{b}{2}\right)^2 - \left(\frac{b}{2}\right)^2 + c$$

1.2. Algebraic Fractions

- An **algebraic fraction** is a fraction that contains **algebraic expressions** in its numerator, denominator, or both
- $\frac{2x+3}{x^2-4}$

Simplifying Algebraic Fractions

- Factorize the numerator and denominator completely
- Cancel out common factors
- $\frac{x^2-9}{x^2-3x} = \frac{(x-3)(x+3)}{x(x-3)} = \frac{x+3}{x}$

Multiplying Algebraic Fractions

- $\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d}$

Dividing Algebraic Fractions

- $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$

Adding and Subtracting Algebraic Fractions

1. Find the **Lowest Common Denominator**
2. Make denominators the same
3. Add or subtract numerators
4. Simplify if possible

$\frac{x}{3} + \frac{2}{x} = \frac{x^2+6}{3x}$

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1.3. Sequences

Linear Sequences

- A linear sequence is a list of numbers where the difference between consecutive terms is constant.
- nth term formula: $T_n = a + (n-1)d$
  - a = first term
  - n = term number
  - d = common difference
- Common Difference: $d = T_{n+1} - T_n$

Quadratic Sequences

- A quadratic sequence has a constant second difference.
- General Form: $T_n = an^2 + bn + c$
  - $2a$ = 2nd Difference (d_2)
  - $3a + b$ = 1st Difference (d_1)
  - $c = T_1 - a - b$

- nth term formula: $T_n = a + (n-1)d_1 + \frac{d_2}{2}(n-1)(n-2)$
  - a = First Term
  - d_1 = First Difference
  - d_2 = Second Difference
- Work out the values and then place them into a formula to work out the nth term formula

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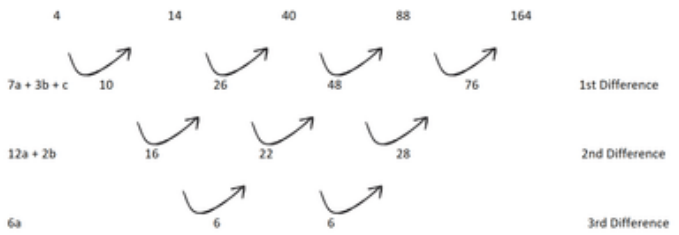
Cubic Sequences

- A cubic sequence has a constant third difference.
- General Form: $T_n = an^3 + bn^2 + cn + d$
  - $7a + 3b + c$ = 1st Difference (d_1)
  - $12a + 2b$ = 2nd Difference ($d_2)
  - $6a$ = 3rd Difference (d_3)
- Find the differences until the third difference is constant.
- Use formulas or simultaneous equations to solve for a, b, c, d.



- nth term formula :
  $T_n = a + (n-1)d_1 + \frac{d_2}{2}(2(n-1))(n-2) + \frac{d_3}{6}(6(n-1))(n-2)(n-3)$
  - d_1 = 1st Difference
  - d_2 = 2nd Difference
  - d_3 = 3rd Difference

Exponential Sequence (Geometric Progression)

- In an exponential sequence, each term is multiplied by a constant ratio r.
- nth term formula = $T_n = ar^{n-1}$
  - a = First Term
  - r = $\frac{T_n}{T_{n-1}}$ (Common Ratio)
  - n = Term Number

1.4. Quadratic Equations

General Form

$ax^2 + bx + c = 0$

where $a \neq 0$

Solving a Quadratic Equation

1. Factorization

- Factorisation means rewriting the quadratic expression into two linear brackets. We look for two numbers that multiply to $a \times c$ and add to b, then solve using the zero-product rule.
- Worked Example

Q1. Solve $x^2 - x - 6 = 0$

Step 1: Identify the coefficients: $a = 1$, $b = -1$, $c = -6$

Step 2: Multiply $a \times c = 1 \times (-6) = -6$

Step 3: Find two numbers that multiply to -6 and add to -1 $\rightarrow$ -3 and 2

Step 4: Factorize: $(x-3)(x+2) = 0$

Step 5: Solve each bracket: $x = 3, -2$

Q2. Solve $2x^2 + 7x + 3 = 0$

Step 1: Identify the coefficients: $a = 2$, $b = 7$, $c = 3$

Step 2: Multiply $a \times c = 2 \times 3 = 6$

Step 3: Find the two terms that multiply to 6 and add to 7 $\rightarrow$ 6 and 1.

Step 4: Rewrite: $2x^2 + 6x + x + 3$

Step 5: Group Items: $2x(x+3) + 1(x+3)$

Step 6: Factorize: $(2x+1)(x+3) = 0$

Step 7: Solve: $x = \frac{-1}{2}, 3$

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2. Quadratic Formula

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$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- When the question says, "Give your answer to two decimal places," use the formula!
- Derivation of the Quadratic Formula is the same as saying "Make x the subject in $ax^2 + bx + c = 0$ "

$$ax^2 + bx + c = 0$$

$$a\left(x^2 + \frac{b}{a}x\right) + c = 0$$

$$a\left(\left(x + \frac{b}{2a}\right)^2 - \frac{b^2}{4a^2}\right) + c = 0 \text{ (Completing the Square)}$$

$$a\left(x + \frac{b}{2a}\right)^2 - \frac{b^2}{4a} + c = 0$$

$$a\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a}$$

$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}$$

$$x + \frac{b}{2a} = \pm\sqrt{\frac{b^2 - 4ac}{4a^2}}$$

$$x + \frac{b}{2a} = \frac{\pm\sqrt{b^2 - 4ac}}{\sqrt{4a^2}}$$

$$x + \frac{b}{2a} = \frac{\pm\sqrt{b^2 - 4\text{ac}}}{2a}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4\text{ac}}}{2a}$$

- **Worked Example**

  Q1. $x^2 - x - 6 = 0$

  Step 1: Identify coefficients

  $a = 1$, $b = -1$, $c = -6$

  Step 2: Substitute in Formula

  $x = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(1)(-6)}}{2(1)}$

  Step 3: Simplify

  $x = \frac{1 \pm \sqrt{1+24}}{2} = \frac{1 \pm \sqrt{25}}{2}$

  Step 4: Solve

  $x = \frac{1+5}{2} = 3$ or $x = \frac{1-5}{2} = -2$

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3. Completing the Square

- Completing the square is a method used to rewrite a quadratic expression of the form $ax^2 + bx + c$ into the form $a(x+p)^2 + q$, which is useful for solving equations and finding the vertex of a parabola.
- If an equation is in completing the square form, $a(x+h)^2 + k$, the coordinates of the turning point (vertex) of the parabola are $(-h, k)$

- **Worked Example**

  Q1. Solve $x^2 + 6x + 5 = 0$

  Step 1: Identify the coefficients

  $a = 1$, $b = 6$, $c = 5$

  Step 2: Take the half of the coefficient of x and square it

  Coefficient of x = 6

  Half of it = $\frac{6}{2} = 3$

  Square it = $3^2 = 9$

  Step 3: Add and subtract this square inside the equation

  $x^2 + 6x + 9 - 9 + 5$

  Step 4: Write as a perfect square

  $(x^2 + 6x + 9) = (x+3)^2$

  Step 5: Solve the equation

  $(x+3)^2 - 4 = 0$

  $(x+3)^2 = 4$

  $x + 3 = \pm 2$

  Step 6: Find the roots

  If $x + 3 = 2$, then $x = -1$

  If $x + 3 = -2$, then $x = -5$

  Step 7: Write the final answer

  $\boxed{x = -1, -5}$

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4. Graphing Method

- Plot the graph of $y = ax^2 + bx + c$

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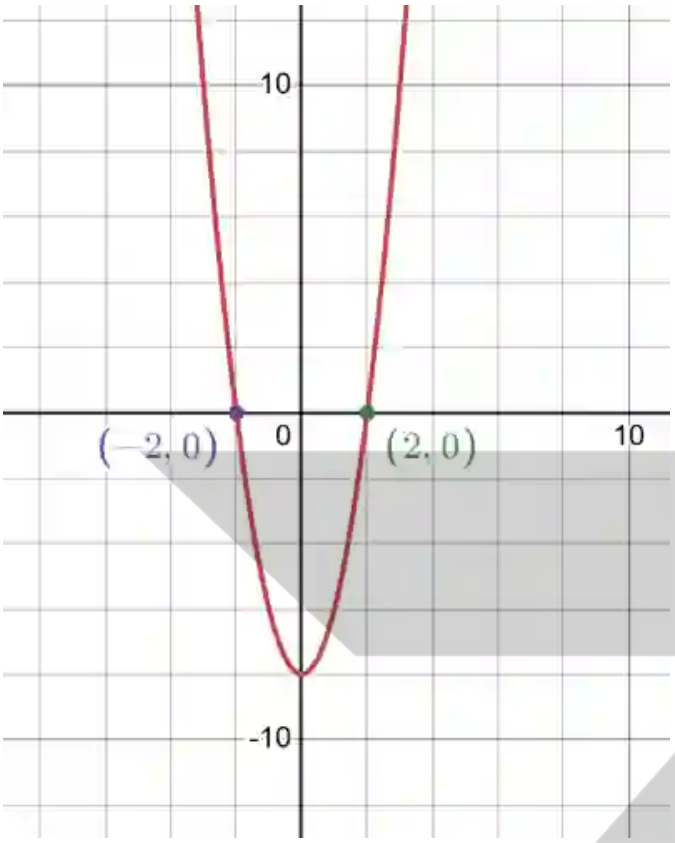
- The points where the graph cuts the x-axis ($y = 0$) are the solutions (roots) of the equation



1.5. Inequalities

- Solve like equations. Perform operations on both sides (add, subtract, multiply, divide) to isolate the variable.
- When you multiply or divide by a negative number, reverse the inequality sign
- **Worked Examples**

  Q1. Solve $y - 3 \geq -7$

  Step 1: Add 3 to both sides

  $y - 3 + 3 \geq -7 + 3$

  $\boxed{y \geq 4}$

  Q2. Solve $\frac{y}{-3} \geq -7$

  Step 1: Multiply both sides by -3

  Because you are multiplying by a negative, reverse the sign

  $y \leq -7 \times -3$

  Step 2: Simplify

  $\boxed{y \leq 21}$

- When two inequalities are present, split them into two
- **Worked Example**

  Q1. Solve $x < 3x - 1 < 2x + 7$

  Step 1: Split into two inequalities

  1. $x < 3x - 1$

  $-2x < -1$

  Divide by -2, flip sign: $x > \frac{1}{2}$

  2. $3x - 1 < 2x + 7$

  $x < 8$

  Step 2: Combine

  $\boxed{\frac{1}{2} < x < 8}$

1.6. Proportion

Direct Proportion

- Two variables, x and y, are **directly proportional** if:

$$y \propto x$$

- This means:

$$y = kx$$

- where k is a non-zero constant, called the constant of proportionality.
- As x increases, y increases and vice versa.
- The ratio $\frac{y}{x} = k$ remains constant

Inverse Proportion

- Two variables, x and y, are **inversely proportional** if:

$$y \propto \frac{1}{x}$$

- This means:

$$y = \frac{k}{x}$$

- where k is a non-zero constant, called the constant of proportionality.
- As x increases, y decreases and vice versa
- The product $xy = k$ remains constant.

Other Types of Proportion

- Linear Proportion: $y \propto x \rightarrow y = kx$
- Square Proportion: $y \propto x^2 \rightarrow y = kx^2$
- Square Root Proportion: $y \propto \sqrt{x} \rightarrow y = k\sqrt{x}$
- Cube Proportion: $y \propto x^3 \rightarrow y = kx^3$
- Cube Root Proportion: $y \propto \sqrt[3]{x} \rightarrow y = k\sqrt[3]{x}$

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1.7. Rectangular Graphs

Standardized Form

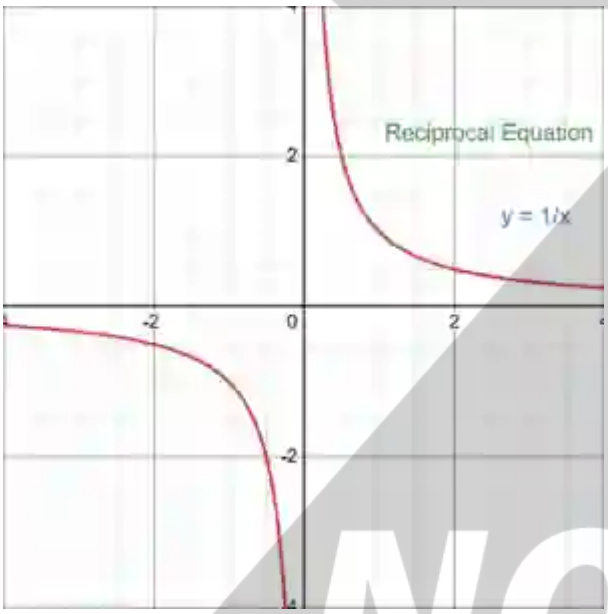
- Basic Form: $y = \frac{a}{x}$
- Transformed Form: $y = \frac{a}{x-h} + k$
  - where (h, k) shifts the hyperbola.

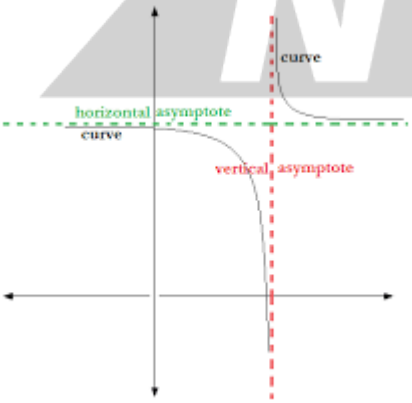
Position of Graph

- If $a > 0$: Graph lies in the **1st** and **3rd** quadrants.
- If $a < 0$: Graph lies in the **2nd** and **4th** quadrants

Asymptotes

- Asymptotes are lines that the curve approaches but never touches.
- For $y = \frac{a}{x-h} + k$:
  - **Vertical Asymptote:** $x = h$
  - **Horizontal Asymptote:** $y = k$
- For Example: For $y = \frac{1}{x}, asymptotesare$ x=0and y=0$





Symmetry

- Hyperbolas are symmetrical about their asymptotes

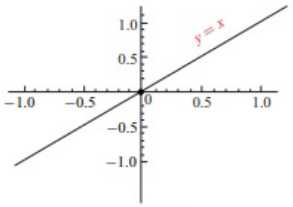


Fig.7.29

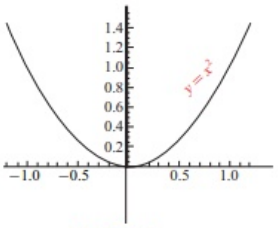


Fig.7.30

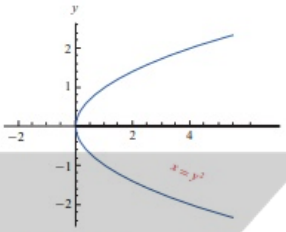


Fig. 7.31

Line of Symmetry for:

Fig 7.29 : y = x

Fig 7.30 : x = 0

Fig 7.31 : y = 0

1.8. Cubic Equation

Standard Form

- $y = ax^3 + bx^2 + cx + d$

Key Properties

- The highest power of x is 3.
- The graph can have:
  - Up to 2 turning points (maxima/minima).
  - One point of inflection (where gradient = 0, but curve doesn't change direction)
- The shape depends on the sign of a:
  - When a > 0, a min-max curve is formed
  - When a < 0 a max-min curve is formed

Turning Points

- Turning points are points on a graph where the gradient (slope) changes sign — the graph changes direction from increasing to decreasing, or vice versa
- Found by differentiating:

$$\frac{dy}{dx} = 3ax^2 + 2bx + c$$

and solving $\frac{dy}{dx} = 0$.

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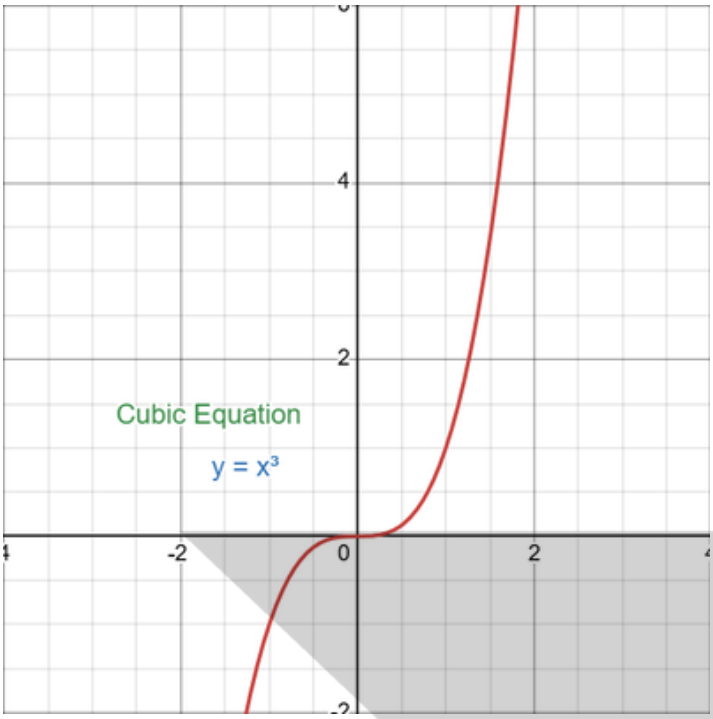

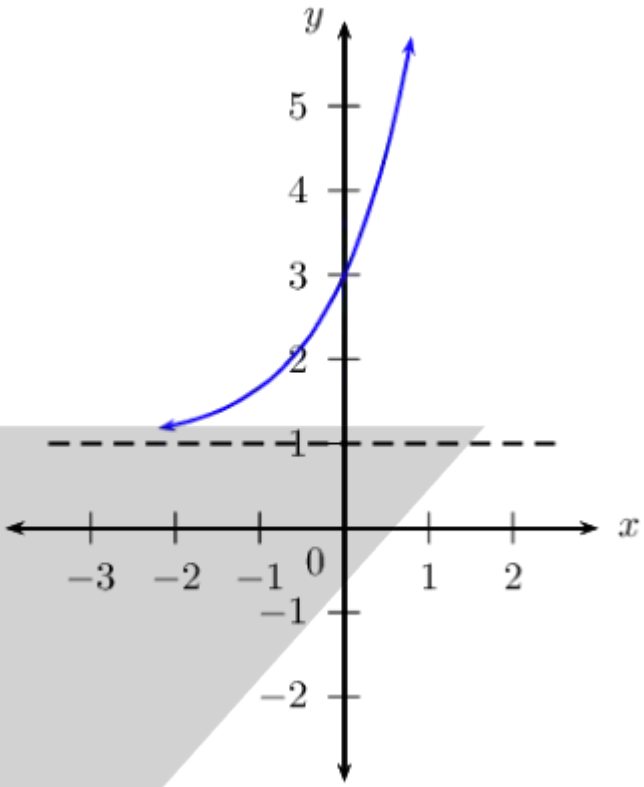

1.9. Exponential Graphs

Standard Form

- $y = a(r^x) + b$

Properties

1. Parameter a

- Represents the **vertical stretch** and determines the initial value when $x = 0$.

2. Parameter r

- Represents the **base** of the exponential function.
- Determines whether the function is increasing or decreasing:
  - If $r > 1$, the curve **increases** as x increases
  - If $0 < r < 1$, the curve **decreases** as x increases

3. Parameter b

- Represents the **vertical shift** of the graph
- Shifts the curve **upwards** or **downward**
- Determines the **horizontal asymptote**: $y = b$

4. Asymptotes

- A curve approaches but never touches these lines
- For $y = a(r^x) + b$, the asymptote is:

  $y = b$
- There is no vertical asymptote, unless the domain is restricted

1.10. Gradient of a Curve

- For a **straight line**, the gradient is **constant** everywhere.
- For a **curve**, the gradient **changes** at different points on the graph.

Method: Finding the Gradient at a Point (By Tangents)

1. Draw the graph
2. Choose the point where you want to find the gradient
3. Draw a tangent to the curve at that point, making sure it just touches the curve at that point (Use a ruler)
4. Find the gradient of the tangent

$$\text{Gradient} = \frac{\text{Change in } y}{\text{Change in } x}$$

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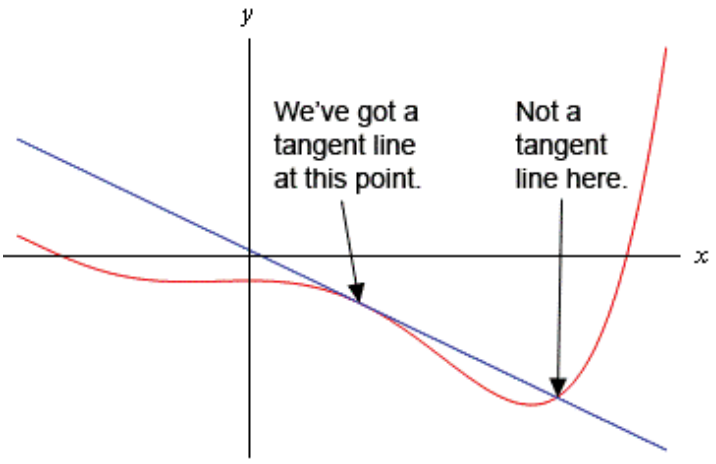

1.11. Differentiation

Understanding Differentiation

- Differentiation finds the **gradient (slope)** of a curve at any point
- It allows us to calculate the gradient of any x-coordinate on a curve.

Power Rule

- For function

  $f(x) = x^n$
- The derivative is:

  $\frac{d}{dx}(x^n) = n \cdot x^{n-1}$
- Steps:
  1. Multiply by the exponent n
  2. Subtract 1 from the exponent
- **Worked Example**

  $\frac{d}{dx}(x^5) = 5x^4$

Differentiating Constants

- The graph is a horizontal line, so the gradient = 0.
- $\frac{d}{dx}(a) = 0$

Stationary Points

- A stationary point occurs where the gradient = 0

  $\frac{dy}{dx}$ = 0
- Steps to find stationary points:
  1. Differentiate the function
  2. Set $\frac{dy}{dx}$ = 0
  3. Solve for x

Nature of Stationary Points

1. **Second Derivative Test**

- Differentiate again to get $\frac{d^2y}{dx^2}$
  - If $\frac{d^2y}{dx^2} > 0$ $\rightarrow$ **Minimum Point**
  - If $\frac{d^2y}{dx^2} < 0$ $\rightarrow$ **Maximum Point**

2. **Gradient Method**

- If the gradient changes $+ \rightarrow -$, the turning point is a **Maximum**
- If the gradient changes from $- \rightarrow +$, the turning point is a **Minimum**

3. **Graph Shape Method**

- Look at the curve around the turning point:
  - If the curve is like a $\cup$, it is a minimum.
  - If the curve is like a $\cap$, it is a maximum.

1.12. Algebraic Indices

- All the laws indices have been covered in the previous chapter. Here, we apply them to solve algebraic questions.

Worked Examples

Example 1

Q. Solve $32^x = 2$

Step 1: Express both sides with the same base

We know: $32 = 2^5$

$32^x = (2^5)^x$

Step 2: Apply the Power Rule

$(2^5)^x = 2^{5x}$

So, the equation becomes: $2^{5x} = 2^1$

Step 3: Use the One-to-One Property of Exponents

$5x = 1$

Step 4: Solve for x

$x = \frac{1}{5}$

Final Answer: $\boxed{x = \frac{1}{5}}$

Example 2

Q. Simplify $3x^{-4} \times \frac{2}{3}x^{\frac{1}{2}}$

Step 1: Group Constants and Variables

$(3 \times \frac{2}{3}) \times (x^{-4} \times x^{\frac{1}{2}})$

Step 2: Simplify Constants

$3 \times \frac{2}{3} = 2$

Step 3: Apply the Product of Powers Rule

Use the law: $x^a \times x^b = x^{a+b}$

$x^{-4} \times x^{\frac{1}{2}} = x^{-4+\frac{1}{2}} = x^{\frac{-7}{2}}$

Step 4: Combine Results

$2 \times x^{\frac{-7}{2}} = 2x^{\frac{-7}{2}}$

Final Answer: $\boxed{2x^{\frac{-7}{2}}}$

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1.13. Simultaneous Equations

Elimination

- Worked Example

Q. Solve $\begin{cases} 3x + 2y = 12 \text{ (Equation 1)} \\ 2x - 2y = 4 \text{ (Equation 2)} \end{cases}$

Step 1: Add both equations (since $+2y$ and $-2y$ will cancel)

$(3x + 2y) + (2x - 2y) = 12 + 4$

$5x = 16 \quad \rightarrow x = \frac{16}{5}$

Step 2: Substitute $x = \frac{16}{5}$ into Equation 2

$2\left(\frac{16}{5}\right) - 2y = 4$

$\frac{32}{5} - 2y = 4$

$-2y = 4 - \frac{32}{5} = \frac{-12}{5}$

$y = \frac{6}{5}$

Final Answer: $\boxed{(x, y) = \left(\frac{16}{5}, \frac{6}{5}\right)}$

Substitution

Example 1 (Linear - Linear)

Q. Solve $\begin{cases} x + y = 10 \text{ (Equation 1)} \\ 2x - y = 4 \text{ (Equation 2)} \end{cases}$

Step 1: From Equation 1,

$x = 10 - y \text{ (Equation 3)}$

Step 2: Substitute into Equation 2

$2(10 - y) - y = 4$

$20 - 2y - y = 4$

$20 - 3y = 4$

$3y = 16$

$y = \frac{16}{3}$

Step 3: Substitute $y = \frac{16}{3}$ into Equation 3

$x = 10 - \frac{16}{3}$

$x = \frac{14}{3}$

Final Answer: $\boxed{(x, y) = \left(\frac{14}{3}, \frac{16}{3}\right)}$

Example 2 (Linear - Quadratic)

Q. Solve $\begin{cases} y = x + 1 \text{ (Equation 1)} \\ y = x^2 + 3x \text{ (Equation 2)} \end{cases}$

Step 1: Equate Equation 1 and Equation 2

$x + 1 = x^2 + 3x$

Step 2: Rearrange

$x^2 + 3x - x - 1 = 0$

$x^2 + 2x - 1 = 0$

Step 3: Solve the quadratic $x = \frac{-2 \pm \sqrt{2^2 - 4(1)(-1)}}{2(1)}$

$x = \frac{-2 \pm \sqrt{4+4}}{2}$

$x = \frac{-2 \pm \sqrt{8}}{2}$

$x = -1 \pm \sqrt{2}$

Step 4: Find the corresponding y

- If $x = -1 + \sqrt{2}$:

  $y = x + 1 = (-1 + \sqrt{2}) + 1 = \sqrt{2}$

- If $x = -1 - \sqrt{2}$:

  $y = x + 1 = (-1 - \sqrt{2}) + 1 = -\sqrt{2}$

Final Answer: $\boxed{(x, y) = \left(-1 + \sqrt{2}, \sqrt{2}\right) \text{ or } \left(-1 - \sqrt{2}, -\sqrt{2}\right)}$

Graphical Method

- For linear + linear:
  - Plot both equations on the same axis
  - The solution is the **point of intersection**
  - If no solution $\rightarrow$ no solution
  - If lines overlap $\rightarrow$ infinitely many solutions
- For linear + quadratic:
  - Intersection = line crosses the curve
  - Number of Intersections tells the number of solutions

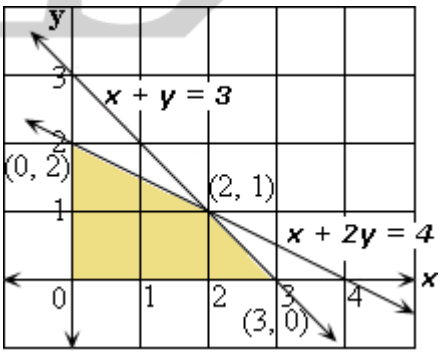
1.14. Linear Programming

Types of Inequalities in a Graph

- **Strict Inequalities** ($<$, $>$) $\rightarrow$ Use a broken/dashed line
- **Non-strict Inequalities** ($\leq$, $\geq$) $\rightarrow$ Use a solid line

Steps to Solve a Linear Programming Problem

- Write inequalities in the form $y = mx + c$
- Draw straight-line graphs of each inequality
  - Use solid or broken lines depending on the inequality type
- Test a point on either side of the line to see which side satisfies the inequality
- Shade the required region for each inequality
  - The common area is called the feasible region



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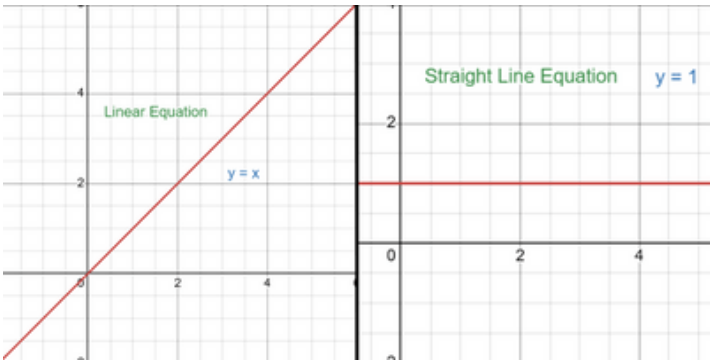


1.15. Functions

Introduction to Functions

- Functions are like pre-set rules: you input a value, apply the rule, and get an output.
- Domain: The set of all possible input values (x-values) for a function.
- Range: The set of all possible output values (y-values) from the function.
- For an **inverse,** the domain and range are swapped

Inverse Function

Steps to find the Inverse Function $f^{-1}(x)$

- **Step 1:** Replace $f(x)$ with y: $y = f(x)$
- **Step 2:** Swap x and y
- **Step 3:** Solve for y in terms of x
- **Step 4:** Replace y with $f^{-1}(x)$

  Example

  - $f(x) = x^2 + 9$
  - $y = x^2 + 9$
  - $x = y^2 + 9$
  - $x - 9 = y^2$
  - $y = \pm\sqrt{x-9}$
  - $f^{-1}(x) = \pm\sqrt{x-9}$

Composite Function

- Notation: $gf(x) = g(f(x))$

Steps to Form a Composite Function

- **Step 1:** Find f(x)
- **Step 2:** Substitute f(x) into g(x)
- **Step 3:** Simplify

- Worked Example

  Find composite function $\begin{cases} f(x) = 3x - 5 \\ g(x) = \frac{3(x+4)}{5} \end{cases}$

  Step 1: Find $f(x)$

  $f(x) = 3x - 5$

  Step 2: Substitute into $g(x)$

  $g(f(x)) = \frac{3((3x-5)+4)}{5}$

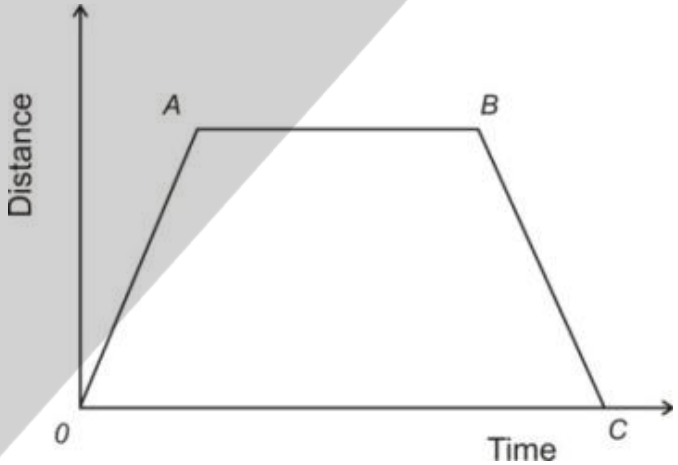
  Step 3: Simplify

  $g(f(x)) = \frac{3(3x-1)}{5} = \frac{9x-3}{5}$

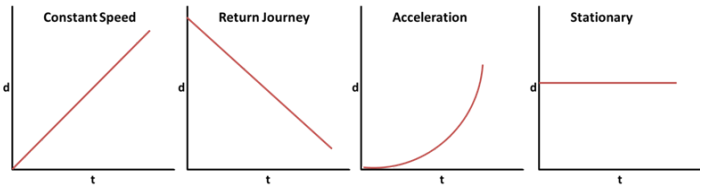
  Final Answer: $\boxed{gf(x) = \frac{9x-3}{5}}$

1.16. Distance-Time Graphs

- $\text{Speed} = \frac{\text{Distance}}{\text{Time}}$
- $\text{Average Speed} = \frac{\text{Total Distance Travelled}}{\text{Total Time Taken}}$



- **From O to A**: Uniform speed
  - Lines with a positive gradient represent objects moving away from the starting point
- **From A to B**: Stationery (speed = 0)
  - Lines with a gradient of 0 represent the object to be at rest
- **From B to C**: Uniform speed (return journey)
  - Lines with a negative gradient represent objects moving towards the starting point



- $\text{Gradient of Distance - Time Graphs} = \text{Speed}$

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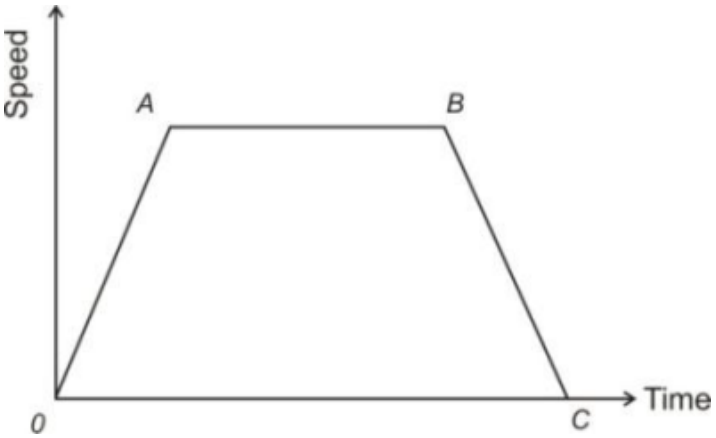
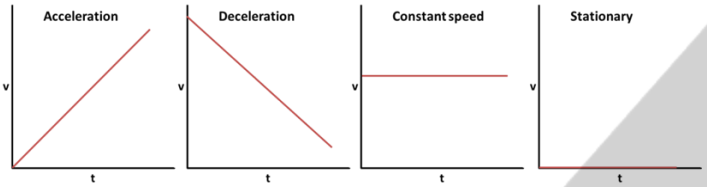
1.17. Speed-Time Graphs

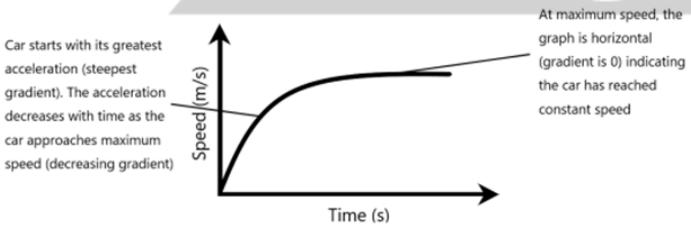


- **From O to A:** Uniform increase in speed
  - Lines with a positive gradient represent constant acceleration
- **From A to B:** Constant speed (acceleration = 0)
  - Lines with 0 gradient represent constant speed
- **From B to C:** Uniform deceleration/retardation
  - Lines with a negative gradient represent deceleration

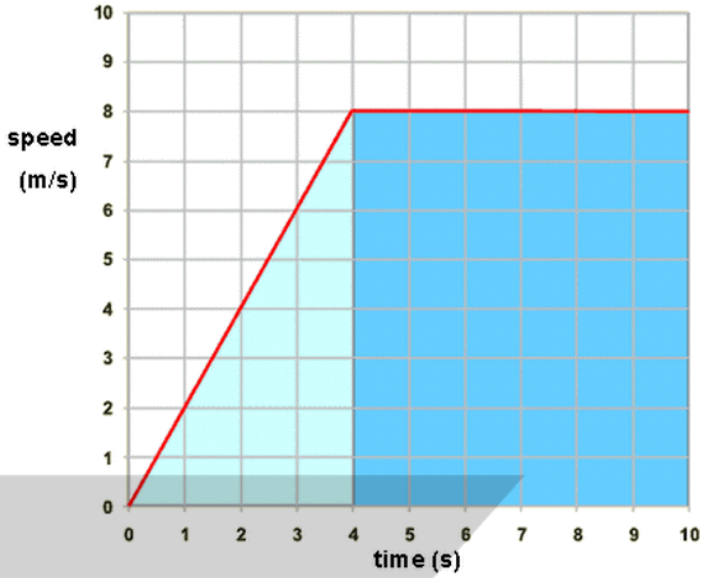


- Graph 1: A Straight Increasing Non-Horizontal Line indicates a constant acceleration where a > 0
- Graph 2: Straight Decreasing Non-Horizontal Line indicates a constant acceleration where a < 0 or deceleration
- Graph 3: Horizontal Line indicates moving at constant speed. a = 0
- Gradient of Speed - Time Graph = Acceleration
- If the acceleration is negative, it is called deceleration or retardation. (The moving body is slowing down.)
- $\text{Acceleration} = \frac{\text{Change in Velocity}}{\text{Time}} = \frac{v-u}{t}$

A car accelerating until it reaches maximum speed



How to find Distance from Speed - Time Graph



Distance = Area under Speed-Time Graph

- $\frac{1}{2} \times 4 \times 8 = 16$
- $6 \times 8 = 48$
- 48 + 16 = 64 meters

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